

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000884	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	23-May-2011
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DG57 / DG517 Service Manual

INSTRUMENT GENERAL OVERVIEW

Procedure description
This document contains an overall description of the instrument, main characteristics modules and block diagrams. For more information see the General Connection Diagram and the Fluidic System Diagram.

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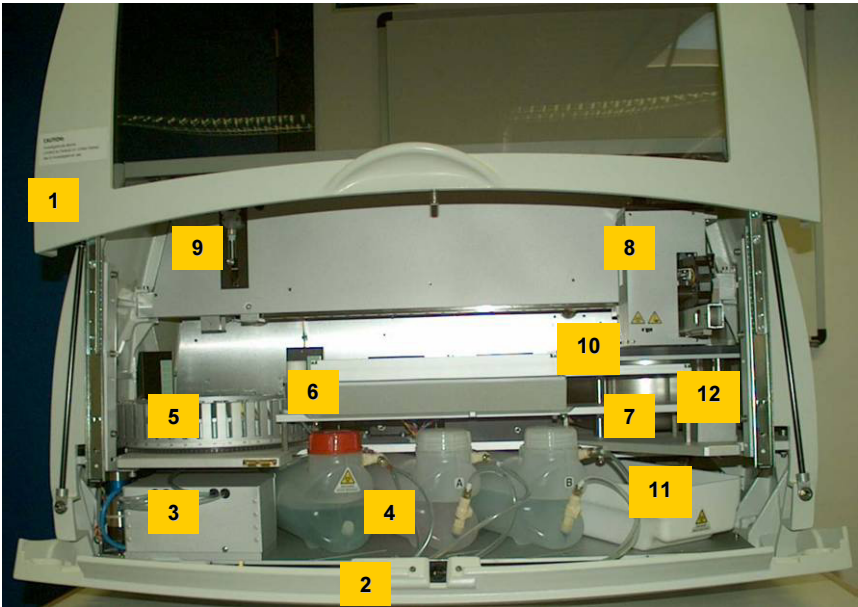
EQUIPMENT

1. ---

SET UP

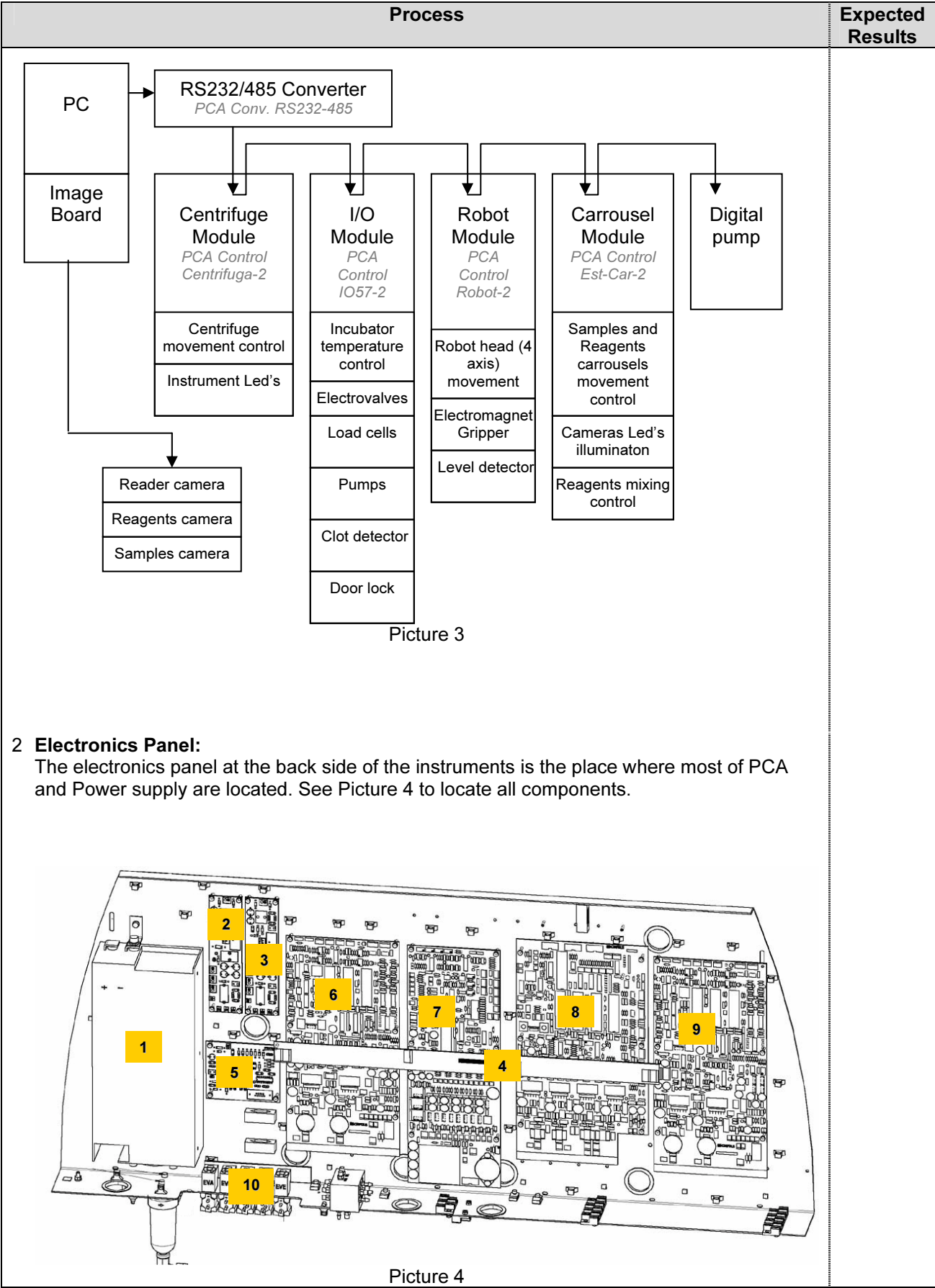
1. ---

PROCEDURE

Process	Expected Results
1 Instrument parts:  Picture 1 1. Main door 2. Bottles door 3. Pumps box 4. Bottles 5. Carrousels 6. Incubator 7. Centrifuge 8. Robot Head 9. Syringe (digital pump) 10. Service Rack 11. Cards waste tray 12. Chute to waste tray	---

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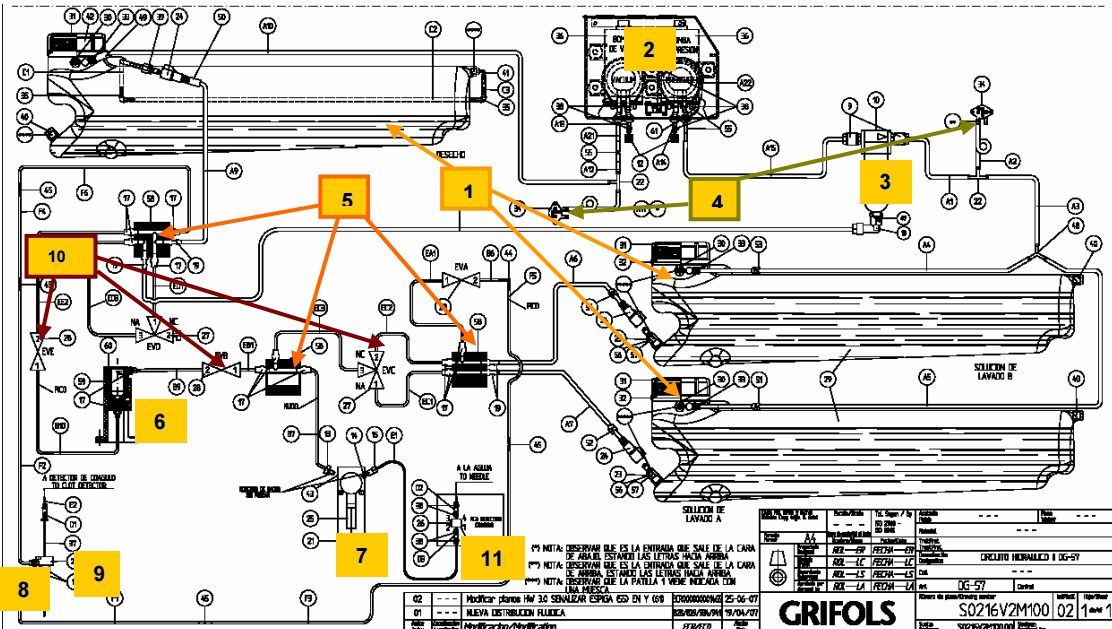
Process	Expected Results
Picture 2 1. Reagents carousel 2. Samples carousel 3. Reagents camera 4. Samples camera 5. Reader camera 6. Centrifuge 7. Dilution station 8. Incubator 9. Service rack 10. Chute to waste tray 11. Chute to waste tray All instrument's functionalities are controlled by several PCA. Each PCA with a microprocessor and all mechanical parts controlled by it are defined as a Module (see Modules structure in Picture 3). The PC communicates to each PCA individually via RS-232 communication. This RS-232 signal is converted to RS-485 standard to connect all PCA among them.	



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Process	Expected Results
1. Power Supply unit 2. PCA CONV 24V/8V 3. PCA CONV 24V/12V 4. PCA POWERBUS 5. PCA CONV. RS232-485 6. PCA CNTR.CENTRIFUGA-2 7. PCA CNTR IO57-2 8. PCA CNTR ROBOT-2 4 EJES 9. PCA CNTR. EST-CAR-2 10. PCA ELECTROVALVES 3 Power Supply: The following diagram (Picture 5) explains the Power Supply distribution. Mains Supply signal is converted into 24VDC. Those 24V are converted to 12V and 8V. NETWORK FILTER (2 T 4A/250V FUSES) POWER-SUPPLY 220-110 V / 24 VDC 300W 24 VDC / 12 VDC PCA Conv 24V/12V 24 VDC / 8 VDC PCA Conv 24V/8V CAMERAS CLOT DETECTOR STATUS LED PCA Conv. RS232-485 PCA POWERBUS PCA Control Centrifuga-2 PCA Control IO57-2 PCA Control Robot-2 PCA Control Est-Car-2 Digital pump (24v only) Picture 5	

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Process	Expected Results
4 Fluidic System: Here are some hints to understand Fluid System general diagram.  Picture 6 1. Bottles (Solution A, Solution B and Waste) 2. Pumps (Pressure and Vacuum) 3. Air filter 4. Pressure and Vacuum sensors 5. Metacrilate Distributor 6. Dilution Cup 7. Digital Pump 8. Probe 9. Wash Station 10. Electrovalves 11. Clot detector	

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	Document reference and PCA's 24V/8V and 24V/12V numbering (step 2) changed.	Document REGD-0000884_v3 replaces document 57001V0M001.
4.0	2010/09/10	It is updated the Picture 6 of fluidic diagram.	Document updated due to the improvements implemented in the fluidic circuit.
5.0	See document header	Updated the Document Change History: Date format changed to YYYY/MM/DD Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
5.0	See document header	Document REGD-0000884_v5 replaces document REGD-0000884_v4	Document version chanced.
5.0	See document header	Document title.	Replaced commercial names by project names.
5.0	See document header	-Pressure regulator removed from picture 4. -Pressure regulator reference removed from point '4. Fluidic system'	Pressure regulation is now performed automatically by pressure pump.
5.0	See document header	Document REGD-0000860 reference deleted.	Document REGD-0000860 not required.
5.0	See document header	Header title.	Increase information to easy document search.